Effects of Intermittent Fasting on Metabolic Markers

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Understanding the Mechanisms Behind Intermittent Fasting

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Intermittent fasting (IF) has gained popularity as a strategy for weight loss and managing obesity-related complications. Studies have consistently shown that IF can improve metabolic markers, including blood glucose levels, cardiac and brain function, and other health benefits, alongside weight loss [2]. However, concerns have also been raised about potential side effects such as muscle loss, ketosis, and electrolyte imbalance. A notable concern is the impact of IF on hormonal circadian rhythms, which have been shown to play a crucial role in normal physiology [2].

Recent research has begun to explore the effects of different modes of intermittent fasting on inflammatory markers in adults [4]. The systematic review conducted by the authors found that various modes of intermittent fasting can influence inflammatory markers, including interleukin (IL)-6, tumor necrosis factor (TNF)α, C-reactive protein (CRP), leptin, and adiponectin. These findings suggest that the mode of fasting may play a significant role in the inflammatory status of individuals.

In the context of reproductive health, intermittent fasting has been studied in relation to hormonal changes in both men and women. In a literature review conducted by our team, we found that intermittent fasting can lead to a decrease in androgen markers, such as testosterone and the free androgen index (FAI), while increasing sex hormone-binding globulin (SHBG) levels in premenopausal females with obesity [3]. This effect was more pronounced when food consumption was confined to earlier in the day. In contrast, fasting did not significantly impact estrogen levels.

Additionally, the menopausal transition, characterized by changes in reproductive hormones, particularly decreased estrogen levels and relative androgen excess, has been observed in adult Muslim women during Ramadan intermittent fasting (RIF) [6]. This study aimed to investigate the effects of RIF on cardiometabolic health markers in premenopausal and postmenopausal healthy women. The results indicated that RIF improved cardiometabolic health markers, with significant improvements in inflammatory markers and lipid

Impact on Metabolic Markers

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Intermittent fasting (IF) has been shown to have significant effects on various metabolic markers, particularly in individuals with prediabetes and type 2 diabetes (T2D). A meta-analysis of randomized trials conducted from the inception to March 2024 found that IF, when compared to a control diet or calorie restriction, leads to improvements in body composition and cardiometabolic health [5]. This meta-analysis included 14 studies involving 1101 adults with prediabetes and T2D, indicating a substantial body of evidence supporting the benefits of IF.

In addition to these broader studies, preliminary research on time-restricted feeding (TRF) suggests that this form of intermittent fasting can improve cardiometabolic health in both rodents and humans [7]. TRF involves fasting for a longer period during the day, with eating restricted to a specific window. A study conducted on overweight adults found that a 4-day randomized crossover study comparing early TRF (eating between 8 am and 2 pm) to a control schedule (eating between 8 am and 8 pm) led to significant improvements in gene expression, circulating hormones, and diurnal patterns in cardiometabolic risk factors [7]. These findings suggest that TRF, a specific form of IF, can have a positive impact on metabolic markers, potentially offering a more practical approach for individuals looking to improve their health through dietary interventions.

Open Questions and Future Research Directions

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Intermittent fasting (IF) has garnered significant attention in recent years due to its potential metabolic benefits. However, the impact of IF on metabolic markers remains a subject of debate and requires further investigation. While some studies suggest that IF can improve metabolic health, others indicate mixed results. For instance, [1] found that intermittent fasting can lead to significant reductions in fasting glucose and insulin levels, which are important markers of metabolic health. However, [2] reported that fasting may not consistently improve other metabolic markers such as triglycerides or HDL cholesterol. These findings highlight the need for more comprehensive studies to understand the full range of metabolic effects of IF. Additionally, the variability in metabolic responses among individuals suggests that personalized approaches to IF may be necessary. Future research should aim to explore the long-term effects of IF on metabolic health, as well as the optimal timing and duration of fasting periods. Furthermore, investigating the mechanisms underlying the observed metabolic changes could provide valuable insights into how IF affects metabolic markers. Lastly, it is crucial to consider the potential

interactions between IF and other lifestyle factors, such as diet and exercise, to better understand the overall impact on metabolic health.

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